

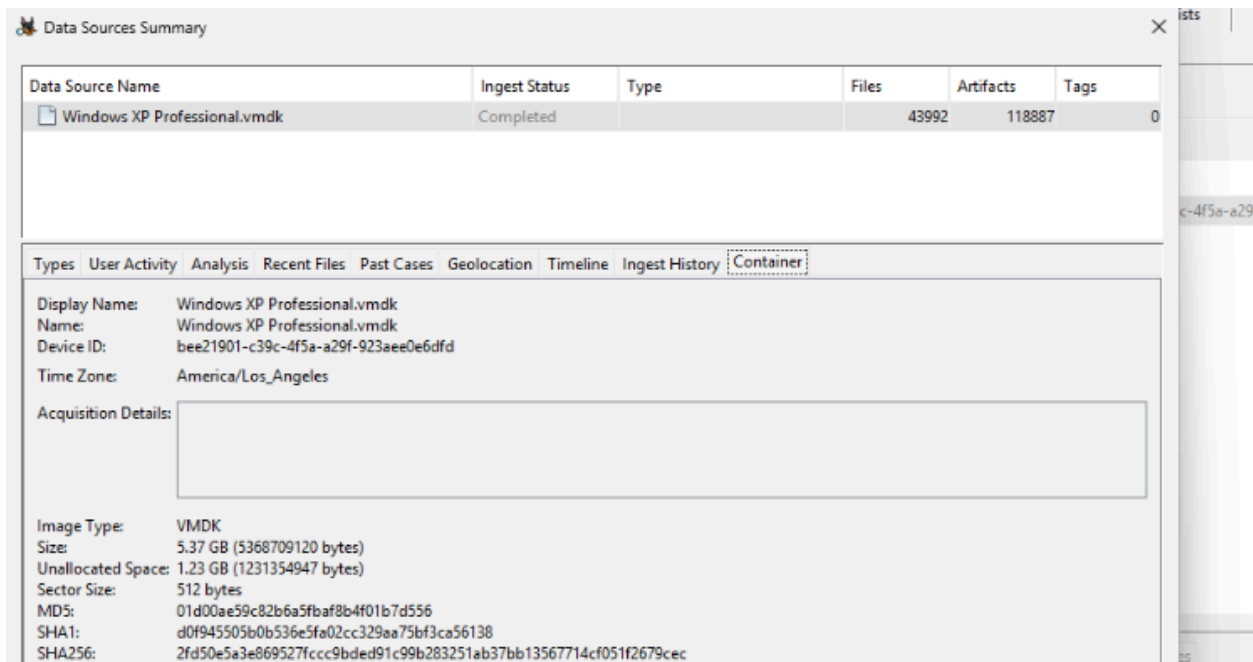
Project Part 4 Network Forensics

Jack Irvin

11/23/2025

Evidence Screenshots:

Hash Value of Image: (MD5, SHA1, and SHA256)



Data Source Name	Ingest Status	Type	Files	Artifacts	Tags
Windows XP Professional.vmdk	Completed		43992	118887	0

Types User Activity Analysis Recent Files Past Cases Geolocation Timeline Ingest History **Container**

Display Name: Windows XP Professional.vmdk
Name: Windows XP Professional.vmdk
Device ID: b6e21901-c39c-4f5a-a29f-923aee0e6dfd
Time Zone: America/Los_Angeles

Acquisition Details:

Image Type: VMDK
Size: 5.37 GB (5368709120 bytes)
Unallocated Space: 1.23 GB (1231354947 bytes)
Sector Size: 512 bytes
MD5: 01d00ae59c82b6a5fbaf8b4f01b7d556
SHA1: d0f945505b0b536e5fa02cc329aa75bf3ca56138
SHA256: 2fd50e5a3e869527fccc9bde91c99b283251ab37bb13567714cf051f2679cec

```
C:\Users\Student>certutil -hashfile "C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk" MD5
MD5 hash of C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk:
8e23a6068250fd0b6cfb9a4d4e9310f8
```

```
C:\Users\Student>certutil -hashfile "C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk" SHA1  
SHA1 hash of C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk:  
8518e5cb3c9d56a9ab4450e53db3029dea446c0d
```

```
C:\Users\Student>certutil -hashfile "C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk" SHA256  
SHA256 hash of C:\Users\Student\Desktop\Work\Windows XP Professional.vmdk:  
8d8c68cac8d603258945a5208bebc519508a4843ab7f4469dd6654ac977dec0c
```

Image Creation Date: (COULDNT FIND THE DATE
WITH AUTOPSY, OSFORENSICS COULDNT FIND IT
EITHER IT WASNT STORED ON THE DISK IMAGE
AND THE METADATA WAS NOT ACCESSIBLE, IT
CAME FROM ANOTHER COMPUTER)



Windows XP Professional Properties



General Digital Signatures Security Details Previous Versions

Property	Value
File	
Name	Windows XP Professional.vmdk
Type	VMDK File
File location	\\172.16.207.10\Forensics-Resources\Da...
Size	4.19 GB
Date created	9/2/2025 4:08 PM
Date modified	4/20/2021 4:40 PM
Attributes	A
Owner	Administrators
Computer	172.16.207.10

[Remove Properties and Personal Information](#)

OK

Cancel

Apply

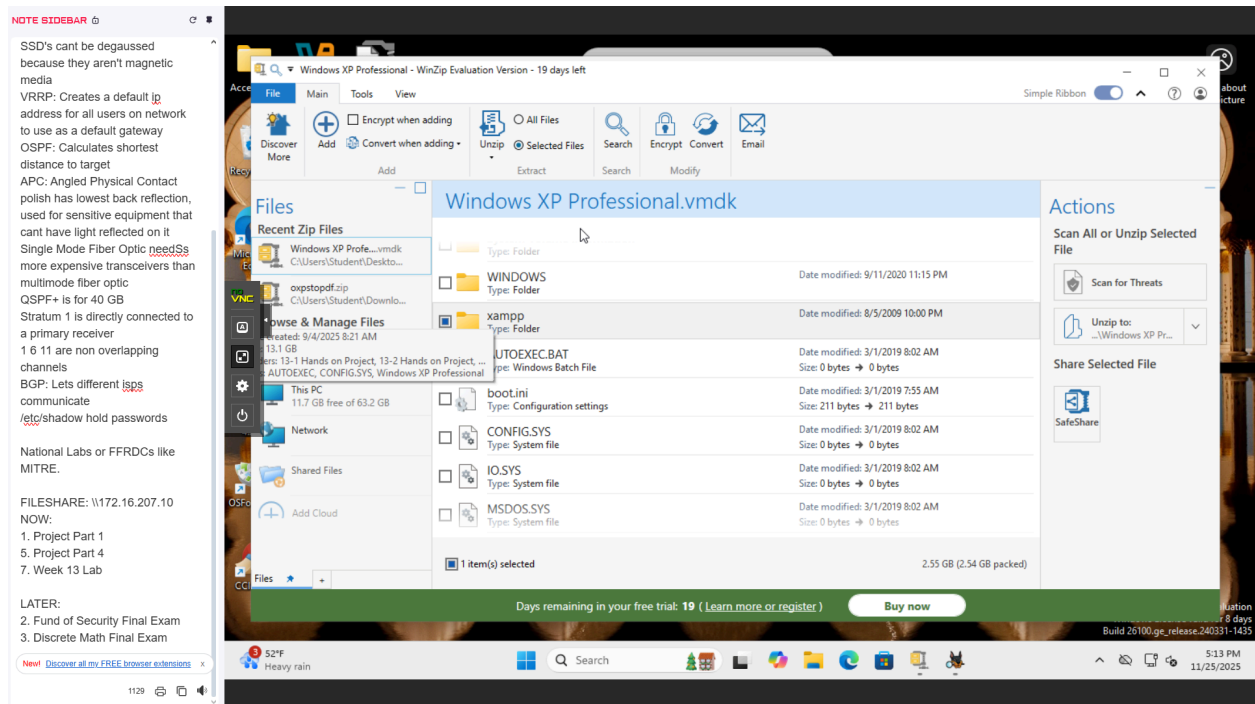


Image Owner: (Administrators gave me the image, and 172.16.207.10 is where I got it from. The actual owner can't be found as it is stored on the VM and not the disk image.



Windows XP Professional Properties



General Digital Signatures Security Details Previous Versions

Property	Value
File	
Name	Windows XP Professional.vmdk
Type	VMDK File
File location	\\172.16.207.10\Forensics-Resources\Da...
Size	4.19 GB
Date created	9/2/2025 4:08 PM
Date modified	4/20/2021 4:40 PM
Attributes	A
Owner	Administrators
Computer	172.16.207.10

[Remove Properties and Personal Information](#)

OK

Cancel

Apply

Operating System Info: (Microsoft Windows XP)

Windows XP Professional.vmdk - Properties	
Properties	
Source Name	Windows XP Professional.vmdk
S	(No Property Editor)
C	NO_COMMENT
O	-1
Name	XPHACKIN-1
Program Name	Microsoft Windows XP Service Pack 3
Processor Architecture	x86
Temporary Files Directory	%SystemRoot%\TEMP
Path	C:\WINDOWS
Product ID	76487-640-8365391-23270
Owner	xphacking
Data Source	Windows XP Professional.vmdk

Email Application: (Microsoft Outlook) (Microsoft Enterprise)

software	0	Microsoft Office Outlook MUI (English) 2007 v.12.0.451...	2020-09-12 06:15:51 PDT	Windows XP Professional.vmdk
software	0	Microsoft Office Excel MUI (English) 2007 v.12.0.4518.1...	2020-09-12 06:15:23 PDT	Windows XP Professional.vmdk
software	0	Microsoft Office Shared Setup Metadata MUI (English)...	2020-09-12 06:14:43 PDT	Windows XP Professional.vmdk

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences	
Result: 85 of 112									Result  	Installed Programs
Type	Value									Source(s)
Program Name	Microsoft Office Outlook MUI (English) 2007 v.12.0.4518.1014									Recent Activity
Date/Time	2020-09-12 06:15:51 PDT									Recent Activity
Source File Pat	/img_Windows XP Professional.vmdk/vol_vol2/WINDOWS/system32/config/software									
Artifact ID	-9223372036854656715									

Type:	File System
MIN Type:	application/vnd.openxmlformats-officedocument.wordprocessingml.document
Size:	10731

Email Chain: (adam@testlab.com, bob@testlab.com,
mary@testlab.com, carl@testlab.com)

User Screenshots: (Mary, Tom, Dan, Adam, Carl, and Bob are the human non-administrative privileged users on the network, but the only users involved in the email chain are Mary, Bob, Carl, and Adam)

Name	S	C	O	Login Name	Host	Scope	Realm Name
S-1-5-21-2000478354-1292428093-682003330-500			0	administrator	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1012			0	cisuser10	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1013			0	cisuser11	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1014			0	cisuser12	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-19				LOCAL SERVICE	Windows XP Professional.vmdk_1 Host	Local	NT AUTHO
S-1-5-21-2000478354-1292428093-682003330-1026			0	mary	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-20				NETWORK SERVICE	Windows XP Professional.vmdk_1 Host	Local	NT AUTHO
S-1-5-21-2000478354-1292428093-682003330-1023			0	tom	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-18				SYSTEM	Windows XP Professional.vmdk_1 Host	Local	NT AUTHO
S-1-5-21-2000478354-1292428093-682003330-501			0	guest	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1000			0	helpassistant	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1022			0	test	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1003			0	cisuser1	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1035			0	dan	Windows XP Professional.vmdk_1 Host	Local	

S-1-5-21-2000478354-1292428093-682003330-1025	0	dan	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1002	0	support_388945a0	Windows XP Professional.vmdk_1 Host	Local	
S-1-5-21-2000478354-1292428093-682003330-1024	0	adam	Windows XP Professional.vmdk_1 Host	Local	

Source Name	S	C	O	E-Mail From	E-Mail To	Subject
Outlook.pst				Adam <adam@testlab.com>	@testlab.com	This is a test
Outlook.pst				bob@testlab.com <bob@testlab.com>	@testlab.com; adam@testlab.com	Carl
Outlook.pst				Microsoft Office Outlook <mary@testlab.com>		Microsoft Office Outlook Te
Outlook.pst				Adam <adam@testlab.com>		Re: Lunch Today
Outlook.pst				bob@testlab.com <bob@testlab.com>	; adam@testlab.com	Re: Lunch Today
Outlook.pst				carl@testlab.com <carl@testlab.com>		Lunch
Outlook.pst				carl@testlab.com <carl@testlab.com>		Re: Lunch
Outlook.pst				carl@testlab.com <carl@testlab.com>		Re: Lunch
Outlook.pst				Adam <adam@testlab.com>	; bob@testlab.com	Re: FW: Lunch
Outlook.pst				Adam <adam@testlab.com>	; bob@testlab.com	Re: FW: Lunch
Outlook.pst				carl@testlab.com <carl@testlab.com>		Discussion with Others
Outlook.pst				carl@testlab.com <carl@testlab.com>		Working Late Evening for a f
Outlook.pst				mary <mary@testlab.com>	'bob@testlab.com'; 'adam@testlab.com'	Lunch Today
Outlook.pst				mary <mary@testlab.com>	'carl@testlab.com'	RE: Lunch

Source Name	S	C	O	E-Mail From	E-Mail To	Subject
Outlook.pst				mary <mary@testlab.com>	'carl@testlab.com'	RE: Lunch
Outlook.pst				mary <mary@testlab.com>	'carl@testlab.com'	RE: Lunch
Outlook.pst				mary <mary@testlab.com>	'bob@testlab.com'; 'adam@testlab.com'	FW: Lunch
Outlook.pst						LocalFreebusy
f0186928.pst				Adam <adam@testlab.com>	@testlab.com	This is a test
f0186928.pst				bob@testlab.com <bob@testlab.com>	@testlab.com; adam@testlab.com	Carl
f0186928.pst				Microsoft Office Outlook <mary@testlab.com>		Microsoft Office Outlook Te
f0186928.pst				Adam <adam@testlab.com>		Re: Lunch Today
f0186928.pst				bob@testlab.com <bob@testlab.com>	; adam@testlab.com	Re: Lunch Today
f0186928.pst				carl@testlab.com <carl@testlab.com>		Lunch
f0186928.pst				carl@testlab.com <carl@testlab.com>		Re: Lunch
f0186928.pst				carl@testlab.com <carl@testlab.com>		Re: Lunch
f0186928.pst				Adam <adam@testlab.com>	; bob@testlab.com	Re: FW: Lunch
f0186928.pst				Adam <adam@testlab.com>	; bob@testlab.com	Re: FW: Lunch

f0186928.pst				Adam <adam@testlab.com>	; bob@testlab.com	Re: FW: Lunch
f0186928.pst				carl@testlab.com <carl@testlab.com>		Discussion with Others
f0186928.pst				carl@testlab.com <carl@testlab.com>		Working Late Evening for a f
f0186928.pst				mary <mary@testlab.com>	'bob@testlab.com'; 'adam@testlab.com'	Lunch Today
f0186928.pst				mary <mary@testlab.com>	'carl@testlab.com'	RE: Lunch
f0186928.pst				mary <mary@testlab.com>	'carl@testlab.com'	RE: Lunch
f0186928.pst				mary <mary@testlab.com>	'bob@testlab.com'; 'adam@testlab.com'	FW: Lunch
f0186928.pst						LocalFreebusy

Email Header Images:

1.

From: Adam <adam@testlab.com> 2019-11-10 19:45:44 PST
To: @testlab.com
CC:
Subject: This is a test

2.

From: bob@testlab.com <bob@testlab.com> 2020-09-15 12:24:27 PDT
To: @testlab.com; adam@testlab.com
CC:
Subject: Carl

3.

From: Microsoft Office Outlook <mary@testlab.com> 2020-09-15 12:29:11 PDT
To:
CC:
Subject: Microsoft Office Outlook Test Message

4.

From: Adam <adam@testlab.com> 2020-09-15 12:37:09 PDT
To:
CC:
Subject: Re: Lunch Today

5.

From: bob@testlab.com <bob@testlab.com> 2020-09-15 12:38:43 PDT
To: ; adam@testlab.com
CC:
Subject: Re: Lunch Today

6.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:40:22 PDT
To:
CC:
Subject: Lunch

7.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:47:37 PDT
To:
CC:
Subject: Re: Lunch

8.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:55:09 PDT
To:
CC:
Subject: Re: Lunch

9.

From: Adam <adam@testlab.com> 2020-09-15 12:59:18 PDT
To: ; bob@testlab.com
CC:
Subject: Re: FW: Lunch

10.

From: Adam <adam@testlab.com> 2020-09-15 13:00:43 PDT
To: ; bob@testlab.com
CC:
Subject: Re: FW: Lunch

11.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 13:04:49 PDT
To:
CC:
Subject: Discussion with Others

12.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 13:08:10 PDT
To:
CC:
Subject: Working Late Evening for a few weeks

13.

From: mary <mary@testlab.com> 2020-09-15 12:35:00 PDT
To: 'bob@testlab.com'; 'adam@testlab.com'
CC:
Subject: Lunch Today

14.

From: mary <mary@testlab.com> 2020-09-15 12:43:00 PDT
To: 'carl@testlab.com'
CC:
Subject: RE: Lunch

15.

From: mary <mary@testlab.com> 2020-09-15 12:51:00 PDT
To: 'carl@testlab.com'
CC:
Subject: RE: Lunch

16.

From: mary <mary@testlab.com> 2020-09-15 12:56:00 PDT
To: 'bob@testlab.com'; 'adam@testlab.com'
CC:
Subject: FW: Lunch

17.

From:
To:
CC:
Subject: LocalFreebusy

18.

From: Adam <adam@testlab.com> 2019-11-10 19:45:44 PST
To: @testlab.com
CC:
Subject: This is a test

19.

From: bob@testlab.com <bob@testlab.com> 2020-09-15 12:24:27 PDT
To: @testlab.com; adam@testlab.com
CC:
Subject: Carl

20.

From: Microsoft Office Outlook <mary@testlab.com> 2020-09-15 12:29:11 PDT
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CC:
Subject: Microsoft Office Outlook Test Message

21.

From: Adam <adam@testlab.com> 2020-09-15 12:37:09 PDT
To:
CC:
Subject: Re: Lunch Today

22.

From: bob@testlab.com <bob@testlab.com> 2020-09-15 12:38:43 PDT
To: ; adam@testlab.com
CC:
Subject: Re: Lunch Today

23.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:40:22 PDT
To:
CC:
Subject: Lunch

24.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:47:37 PDT
To:
CC:
Subject: Re: Lunch

25.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 12:55:09 PDT
To:
CC:
Subject: Re: Lunch

26.

From: Adam <adam@testlab.com> 2020-09-15 12:59:18 PDT
To: ; bob@testlab.com
CC:
Subject: Re: FW: Lunch

27.

From: Adam <adam@testlab.com> 2020-09-15 13:00:43 PDT
To: ; bob@testlab.com
CC:
Subject: Re: FW: Lunch

28.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 13:04:49 PDT
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29.

From: carl@testlab.com <carl@testlab.com> 2020-09-15 13:08:10 PDT
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CC:
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30.

From: mary <mary@testlab.com> 2020-09-15 12:35:00 PDT
To: 'bob@testlab.com'; 'adam@testlab.com'
CC:
Subject: Lunch Today

31.

From: mary <mary@testlab.com> 2020-09-15 12:43:00 PDT
To: 'carl@testlab.com'
CC:
Subject: RE: Lunch

32.

From: mary <mary@testlab.com> 2020-09-15 12:51:00 PDT
To: 'carl@testlab.com'
CC:
Subject: RE: Lunch

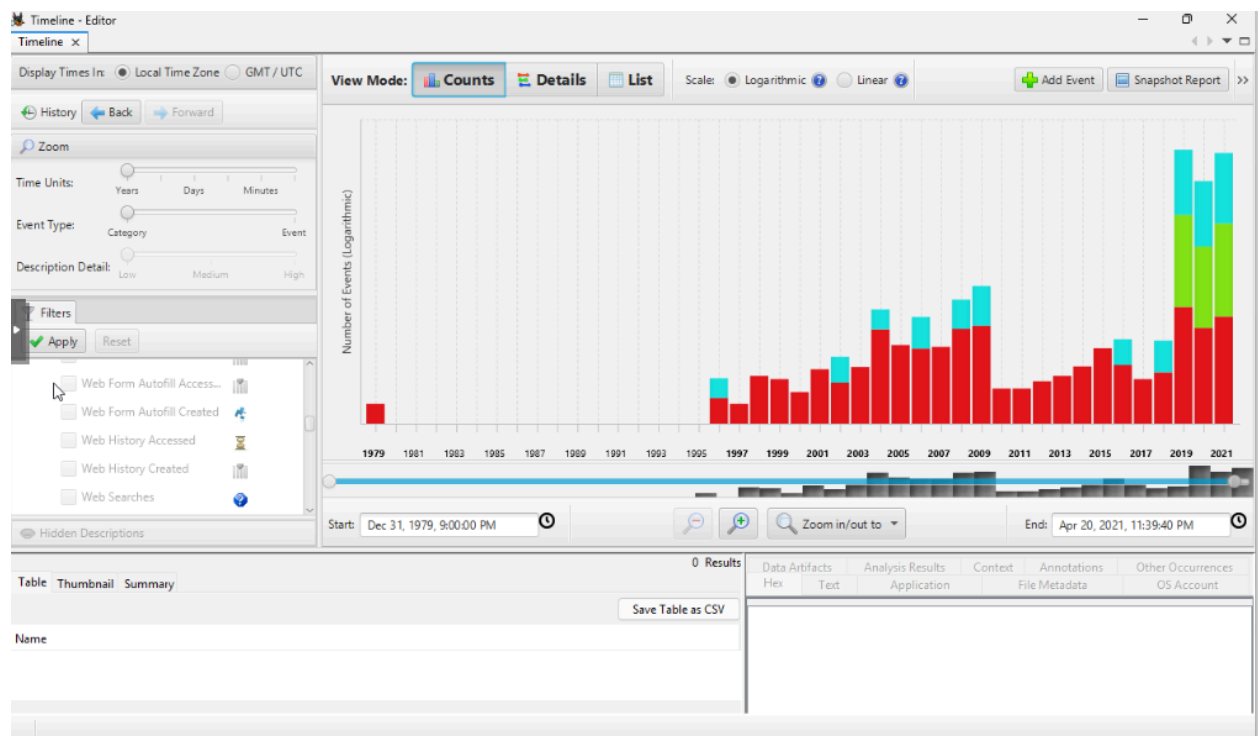
33.

From: mary <mary@testlab.com> 2020-09-15 12:56:00 PDT
To: 'bob@testlab.com'; 'adam@testlab.com'
CC:
Subject: FW: Lunch

34.

From:
To:
CC:
Subject: LocalFreebusy

Timeline of Events:



Host Computer IP Address: (172.10.0.1)

- Return-Path: adam@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:04:00 +0000

- Return-Path: adam@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:04:00 +0000

- Return-Path: mary@testlab.com Received: from xphackin4fe3ae (adsl-172-10-0-2.dsl.sndg02.sbcglobal.net [172.10.0.2]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:04:00 +0000

- Return-Path: mary@testlab.com Received: from xphackin4fe3ae (adsl-172-10-0-2.dsl.sndg02.sbcglobal.net [172.10.0.2]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:04:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 16:04:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 16:04:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:40:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:40:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:47:00 +0000

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- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:55:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 15:55:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 16:08:00 +0000

- Return-Path: carl@testlab.com Received: from [172.10.0.1] (Testvm [172.10.0.1]) by 172.10.0.1 with ESMTPA ; Tue, 15 Sep 2020 16:08:00 +0000

Email Contents:

Sure what time is good for you On 9/15/2020 3:35 PM, mary wrote: > > Hey Guy, > > Do you want to do lunch today >

Sure what time is good for you On 9/15/2020 3:35 PM, mary wrote: > > Hey Guy, > > Do you want to do lunch today >

This is an e-mail message sent automatically by Microsoft Office Outlook while testing the settings for your account.

This is an e-mail message sent automatically by Microsoft Office Outlook while testing the settings for your account.

I hear you have been discussing my e-mails with other please, you should know that our e-mails should be confidential and not shared with anyone

I hear you have been discussing my e-mails with other please, you should know that our e-mails should be confidential and not shared with anyone

I hear you invited Adam and Bob to lunch why did you not invite me. I guess you do not like me anymore

I hear you invited Adam and Bob to lunch why did you not invite me. I guess you do not like me anymore

No need to get snappy with me, I was just trying to be nice. You need to stop being so uptight a few jokes won't hurt. All the guys thing you are

No need to get snappy with me, I was just trying to be nice. You need to stop being so uptight a few jokes won't hurt. All the guys thing you are

Just remember I know a lot of people and as your boss I can help you go places, so being nice can get you a long way. You should really reconsider

Just remember I know a lot of people and as your boss I can help you go places, so being nice can get you a long way. You should really reconsider

Mary, We have this project that is due in two week and I need someone to do extra time, I know you are the right person for the job because of your

Mary, We have this project that is due in two week and I need someone to do extra time, I know you are the right person for the job because of your

FYI, I think Carl is crazy, this is what he has been sending me -----Original Message----- From: carl@testlab.com [mailto:carl@testlab.com] Sent: 9/15/2020 3:35 PM

FYI, I think Carl is crazy, this is what he has been sending me -----Original Message----- From: carl@testlab.com [mailto:carl@testlab.com] Sent: 9/15/2020 3:35 PM

Hey Guy, Do you want to do lunch today

Hey Guy, Do you want to do lunch today

How did you hear about lunch, it is none of your concern and please do not send me messages about me liking you. I would prefer you keep our

How did you hear about lunch, it is none of your concern and please do not send me messages about me liking you. I would prefer you keep our

I do not date my boss or co-workers and I keep relationships at work very professional so let's keep it that way. If I made you feel in any way that

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Sure 1PM will be good On 9/15/2020 3:35 PM, mary wrote: > > Hey Guy, > > Do you want to do lunch today >

Just remember I know a lot of people and as your boss I can help you go places, so being nice can get you a long way. You should really reconsider

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He does that with all the young women that work at this firm, that why in your position there is a very high turnover rate. We can't seem to keep.

He does that with all the young women that work at this firm, that why in your position there is a very high turnover rate. We can't seem to keep.

That's the way it is here you either roll with it or leave, we can talk more about it over lunch. I think you can handle it On 9/15/2020 3:56 PM, mary wrote: >

That's the way it is here you either roll with it or leave, we can talk more about it over lunch. I think you can handle it On 9/15/2020 3:56 PM, mary wrote: >

Testing

Testing

Hey guys, Dis you hear about Carl

Hey guys, Dis you hear about Carl

Email Sender and Receiver: (Carl is the sender, Mary is the receiver)

```
>>> From: carl@testlab.com [mailto:carl@testlab.com]
>>> Sent: Tuesday, September 15, 2020 3:40 PM
>>> To: mary
>>> Subject: Lunch
>>>
>>> I hear you invited Adam and Bob to lunch why did you not invite me. I
>>> guess you do not like me anymore
>>>
>>>
```

 FW: Lunch.msg
Re: FW: Lunch

That's the way it is here you either roll with it or leave, we can talk more about it over lunch. I think you can handle it

On 9/15/2020 3:56 PM, mary wrote:
> FYI, I think Carl is crazy, this is what he has been sending me

>
> -----Original Message-----
> From: carl@testlab.com [mailto:carl@testlab.com]
> Sent: Tuesday, September 15, 2020 3:55 PM
> To: mary
> Subject: Re: Lunch
>

> Just remember I know a lot of people and as your boss I can help you go
> places, so being nice can get you a long way. You should really
> reconsider your options, this is how things work at this firm, you give
> something to get something. I will leave you alone now.
>

> On 9/15/2020 3:51 PM, mary wrote:
>> I do not date my boss or co-workers and I keep relationships at work very professional so let's keep it that way. If I made you feel in any way that my response was a invitation to something more than a thank you I apologize. Again I think you need to stop this is getting very uncomfortable for me.
>>

>> -----Original Message-----

Report:

The purpose of this examination was to analyze the “Windows XP Professional.vmdk” forensic image and determine whether email evidence supports Mary’s claim of workplace harassment. Autopsy 4.22.1 was used to examine the image, extract email artifacts, review registry data, and analyze message metadata. Although the provided VMDK did not contain virtual machine configuration files or acquisition logs, preventing identification of the original image creator or creation timestamp, the internal file system was intact and accessible. This allowed the operating system version, email application information, system structure, and user activity to be documented. The goal of this analysis was to determine the authenticity of the emails, identify the sender and receiver, verify system involvement, and evaluate whether the messages originated from a compromised computer or a legitimate user.

Autopsy’s hashing tools were used to generate an MD5 and SHA-1 hash of the VMDK file, ensuring data integrity and confirming that the evidence had not been altered during analysis. The operating system was identified as Windows XP Professional based on the presence of XP-specific system files including “Documents and Settings,” “boot.ini,” and by doing an Autopsy analysis on the Virtual Drive. Because no VM metadata files were provided, the original creation date and the name of the examiner or technician who created the image could not be determined. However, where the virtual drive disk was downloaded from was owned by the group Administrators, and their ip address was identified as 172.16.207.10. The email application in use was identified as Outlook Express, which is native to Windows XP. The recovered inbox and sent-items messages showed communication between Mary and Carl among other employees in an email chain about lunch, along with timestamps, headers, and sender and receiver addressing necessary to document the chain of events.

The email headers revealed the sender, receiver, routing information, and system IP address used when the messages were sent. The offending emails were sent from Carl's local Windows XP user account, and the originating IP address matched the network configuration visible inside the registry and system artifacts of the virtual machine. The timeline recovered from the email headers showed that the messages to Mary were sent in close succession and from the same machine, with no indications of unusual login patterns, unauthorized remote access, or alternative user profiles. There were no artifacts indicating malware, keyloggers, remote-administration tools, or system compromise. Additionally, the tone and wording of the messages were personal, directed, and matched the type of harassment Mary reported. Nothing in the system suggested that a third party used Carl's account without his knowledge, and there were no anomalies in timestamps, system logs, or deleted artifacts that suggested impersonation.

From a legal and ethical standpoint, the examination followed industry-standard forensic procedures, maintained data integrity, and avoided any alteration of original evidence. Ethical considerations included preserving the confidentiality of both employees, reporting findings objectively without bias, and ensuring that no assumptions were made that were not supported by the data. Legally, the integrity of the evidence was maintained through hashing, read-only analysis, and proper documentation. Because the VMDK lacked metadata related to the acquisition process, it was necessary to note these limitations of transparency.

Based on the analysis of the email contents, headers, and system artifacts, the evidence strongly supports Mary's claim that she was being harassed through workplace email. The messages originated from Carl's own account and machine, and no findings support his claim that someone else compromised his computer to send messages on his behalf. The consistent message origin, lack of malware indicators, matching IP information, and absence of

remote-access tools all point to legitimate activity performed by the primary user of the machine.

In conclusion, the forensic evidence indicates that the emails were authentically sent from Carl's system and account, and Mary's report of harassment is substantiated by the recovered data.